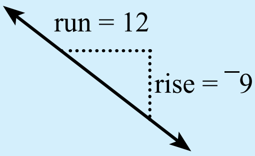
Gradient

The gradient, or slope, or m, is used to describe the steepness of lines.

$$m = \text{gradient} = \frac{\text{rise}}{\text{run}} = \frac{\text{change in y}}{\text{change in x}}$$

Exercise 16.1

Find the gradient of each of the following line segments:

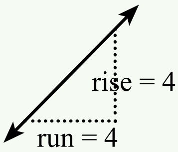


$$\text{Gradient} = m = \frac{\text{rise}}{\text{run}}$$

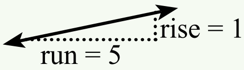
$$m = \frac{^{-}9}{12}$$

$$m = {}^{-}0.75$$

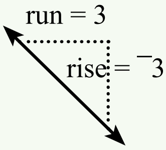
1



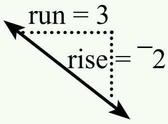
2

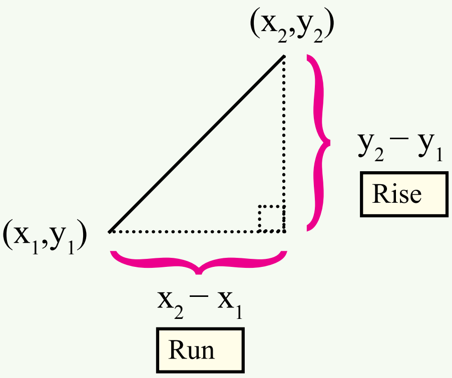


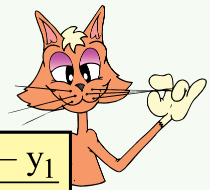
3



4

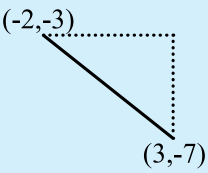






$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Slopes up to the right ▶ positive gradient
Slopes down to the right ▶ negative gradient



First complete:

$x_1 = -2$, $y_1 = -3$
$x_2 = 3$, $y_2 = -7$

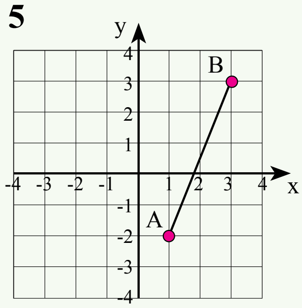
$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{^{-}7 - {}^{-}3}{3 - {}^{-}2}$$

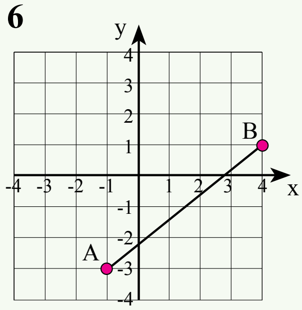
$$m = \frac{^{-}4}{5}$$

$$m = {}^{-}0.8$$

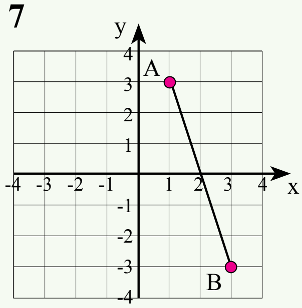
5



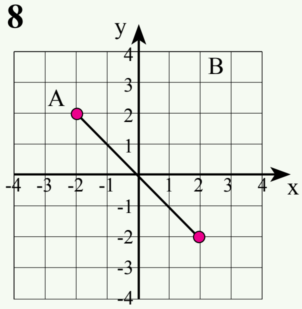
6



7



8



A(-1,4) and B(2,-3)

First complete: $x_1 = -1, y_1 = 4$
$x_2 = 2, \quad y_2 = -3$

Gradient = $m = \dfrac{y_2 - y_1}{x_2 - x_1}$

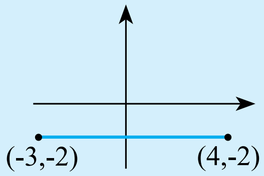
$m = \dfrac{(-3-4)}{(2--1)}$

$m = -2.33$

9	A(1,1), B(3,4)	**10**	A(2,5), B(6,1)	**11**	A(5,2), B(7,5)
12	A(2,3), B(4,1)	**13**	A(4,1), B(-2,3)	**14**	A(2,3), B(-5,-1)
15	A(3,0), B(-7,-8)	**16**	A(-3,1), B(-2,4)	**17**	A(-1,-2), B(3,1)

Exercise 16.2

Find the gradient of each of the following line segments and decide whether the line segment is parallel to the x-axis or the y-axis:



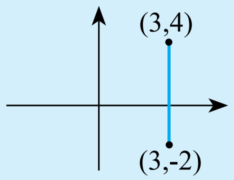
$x_1 = 4, y_1 = -2$
$x_2 = -3, y_2 = -2$

Gradient = $m = \dfrac{y_2 - y_1}{x_2 - x_1}$

$m = \dfrac{(-2--2)}{(-3-4)}$

$m = \dfrac{0}{-7} = 0$

The line is parallel to the x-axis.



$x_1 = 3, y_1 = 4$
$x_2 = 3, y_2 = -2$

Gradient = $m = \dfrac{y_2 - y_1}{x_2 - x_1}$

$m = \dfrac{(4--2)}{(3-3)}$

$m = \dfrac{6}{0} = \infty$

The line is parallel to the y-axis.

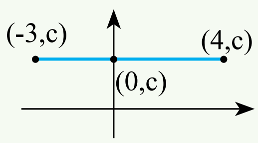
1	A(2,1), B(5,1)	**2**	A(5,3), B(6,3)	**3**	A(1,-2), B(7,-2)
4	A(2,-3), B(2,1)	**5**	A(4,1), B(4,-5)	**6**	A(-2,3), B(-2,-1)
7	A(3,0), B(-7,0)	**8**	A(-3,1), B(-3,4)	**9**	A(-1,-2), B(3,-2)

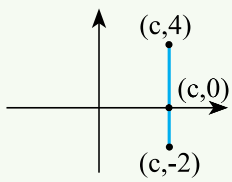
Gradient of lines parallel to the x-axis:

$$\mathbf{m = 0}$$

Gradient of lines parallel to the y-axis:

$$\mathbf{m = \infty}$$





Equation of a line parallel to the x-axis:

$$\mathbf{y = c}$$

Equation of a line parallel to the y-axis:

$$\mathbf{x = c}$$